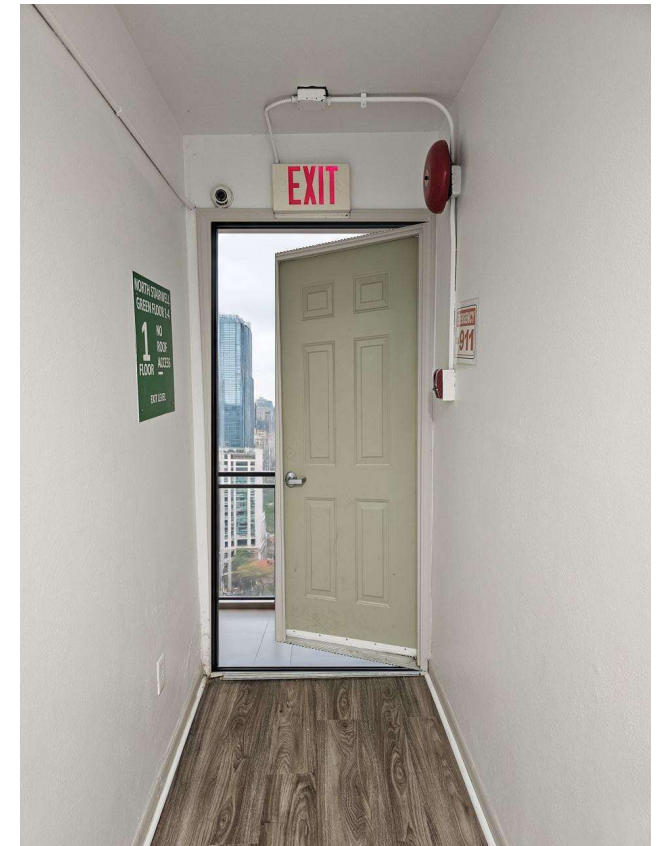


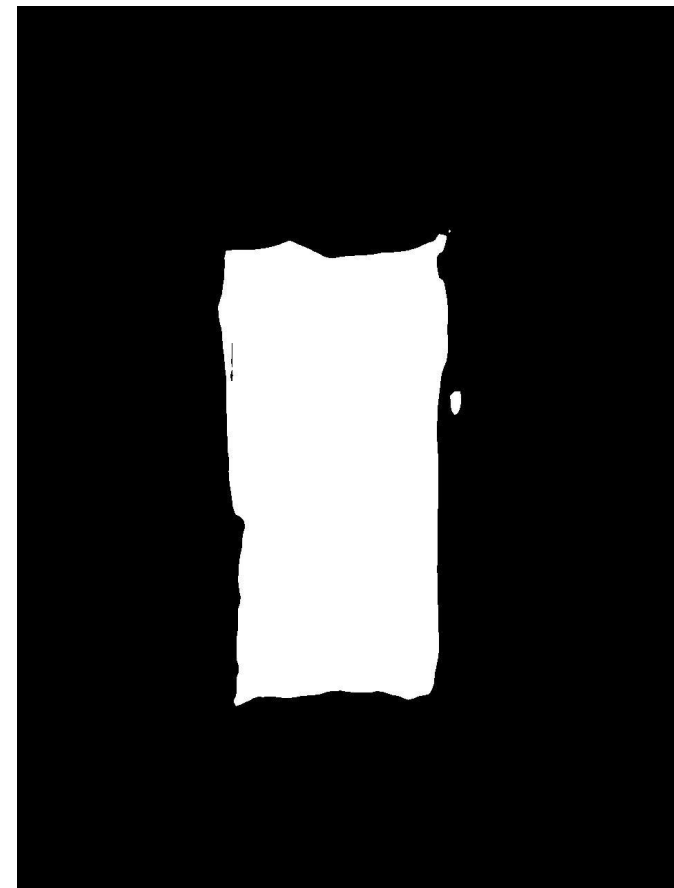
2D DRAWER Implementation

Hacker: Casey Rodgers

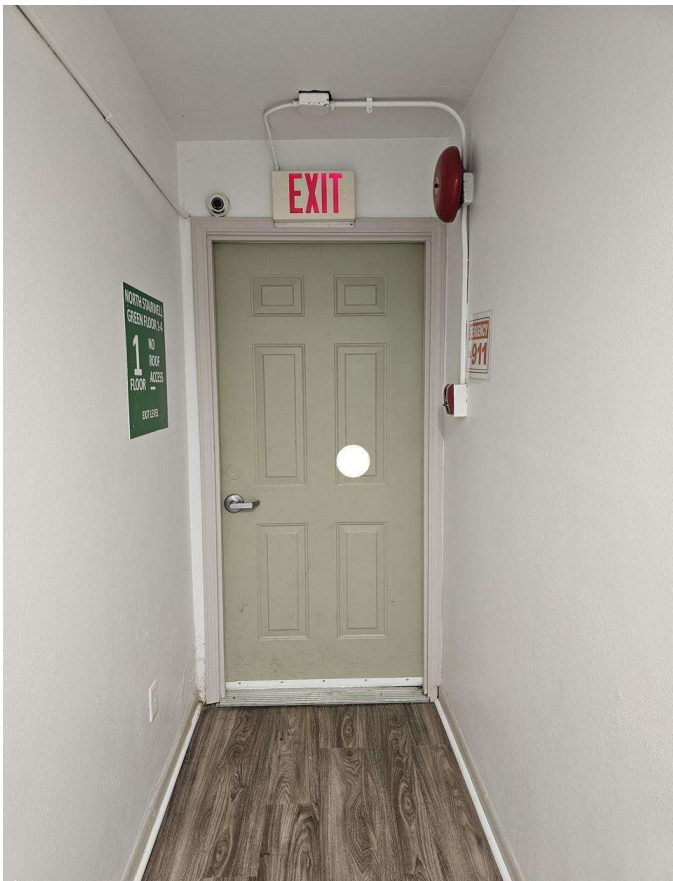
Photo to Openable Objects?



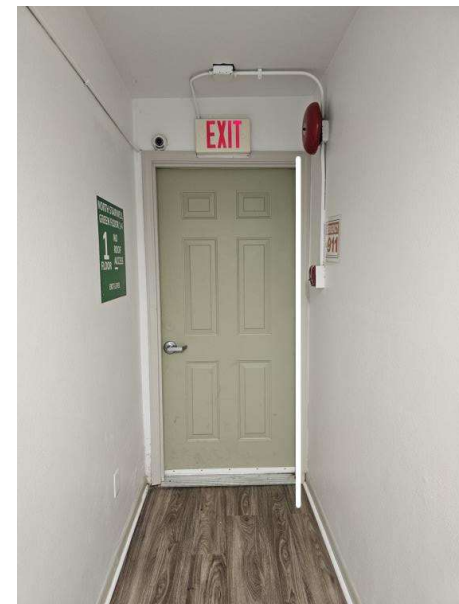
1.) Detect the Openable Objects



2.) Determine Object Affordances



Movable: one_hand
Rigid: yes
Kinematics: rotation
Action: pull



3.) Extract object cleanly



4.) Warp ROI



OpenCV + Homography



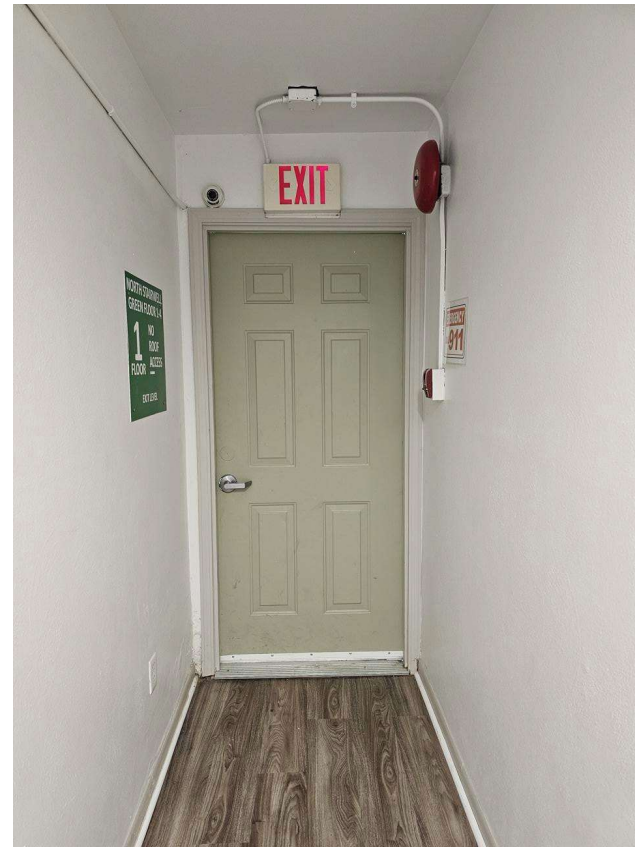
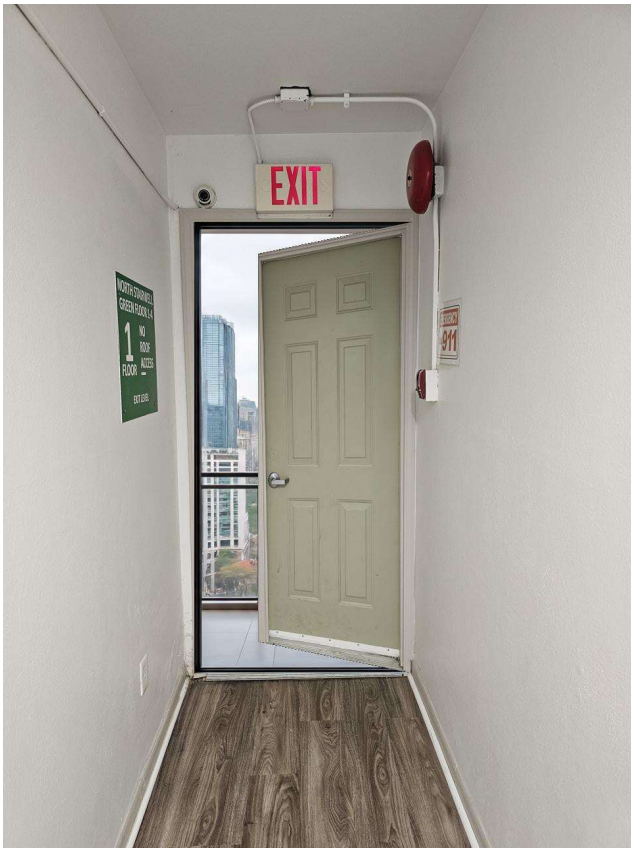
5.) Create background based on object

"Can you generate an image of the most likely image that would be behind an apartment door looking outside the apartment?"

Llama4



6.) Combine Together with OpenCV



Exploration

- Originally tried Colab -> Weird file structure errors -> Local Jupyter Notebook
- Spent a lot of time trying to:
 - Get the model packages to agree
 - Get the inputs in the right format
 - Fix minor errors in between each model
 - Playing with different inputs and settings
- Originally tried to directly incorporate a stable diffusion model -> Super slow -> Used an external diffusion model

Lessons Learned / Reflection

- Combining existing models into one workflow is difficult
 - Each one requires inputs in a certain format
 - Each one produces outputs in a certain format
 - Each one requires different package versions
 - Not straight forward to use only certain functions
- Beware converting from Pillow images to Numpy to OpenCV
- Each model has errors -> These errors propagate